

4. Run Your Program:

- Open `index.html` in your web browser.
- Click the button and see the text change dynamically.

JavaScript in Web Development

JavaScript is a core technology of the web, alongside HTML and CSS. Here are some key aspects of how JavaScript enhances web development even to this day:

1) DOM Manipulation:

JavaScript can dynamically update the content, structure, and style of web pages by interacting with the DOM.

Example: Changing the background color of a webpage

```
document.body.style.backgroundColor = "lightblue";
```

2) Event Handling:

JavaScript handles user interactions through events like clicks, key presses, and form submissions.

Example: Handling a form submission

```
const form = document.querySelector('form');
form.addEventListener('submit',
function(event) {
    event.preventDefault();
    alert("Form submitted!");
});
```

3) Asynchronous Programming:

JavaScript supports asynchronous programming, allowing you to perform tasks like fetching data from an API without blocking the main thread.

Example: Fetching data from an API

```
fetch('https://api.example.com/data')
    .then(response => response.json())
    .then(data => console.log(data))
    .catch(error => console.error('Error:',
error));
```

JavaScript Frameworks and Libraries

JavaScript's capabilities are extended by a wide range of frameworks and libraries, which simplify development and provide powerful tools.

Popular Frameworks:**1. React:**

- A library for building user interfaces, developed by Facebook.
- Uses a component-based architecture.
- **Example:**

```
// Basic React component
function App() {
    return <h1>Hello, World!</h1>;
}
```

```
// Render the component
ReactDOM.render(<App />, document.
getElementById('root'));
```

2. Vue.js:

- A progressive framework for building user interfaces.
- Easy to integrate into projects and scale between a library and a full framework.
- **Example:**

```
<!-- Vue.js component -->
<div id="app">{{ message }}</div>
<script>

    new Vue({
        el: '#app',
        data: {
            message:

'Hello, Vue!'

        }
    });

</script>
```

3. Angular:

- A platform for building mobile and desktop web applications, developed by Google.

